ToxCast Assay Prioritization for Developmental and Reproductive Toxicity Screening

Hanyi Seok

Research at University of Seoul | Jun 2021–Feb 2022 | Supervisor: Prof. Jinhee Choi

Objective and contribution

This project linked ToxCast in vitro assay activity with rat developmental and reproductive study records in ToxRefDB to prioritize candidates for follow-up screening. It explored how existing animal-study data could inform the selection of in vitro assays that might complement animal testing.

Working within the laboratory team, I integrated records spanning 449 chemicals and 949 assays, matched chemicals by CAS number, distinguished inactive results from missing observations, and performed endpoint-specific analyses in R. These decisions aligned assay activity with the available ToxRef values while retaining the number of paired chemicals for each comparison.

Data integration and statistical analysis

Rat prenatal developmental studies (EPA guideline 870.3700) and multigenerational reproductive studies (870.3800) were analyzed separately; mouse data were too sparse for this analysis. Records with missing values and ToxRef entries marked N were excluded within each comparison. ToxCast activity was encoded as $\text{active} = 1$ and $\text{inactive} = 0$, and retained numeric ToxRef values were $\log_{10}$-transformed. Pearson correlation between binary activity and the continuous transformed endpoint is equivalent to point-biserial correlation, matching the two variable types. Pairwise chemical counts were retained to show the support for each association.

Candidate prioritization

Assays with an absolute correlation of at least 0.5 and at least 15 paired chemicals were shortlisted. Four met these criteria for each study type. The assay with the largest positive correlation in each type was then reviewed biologically.

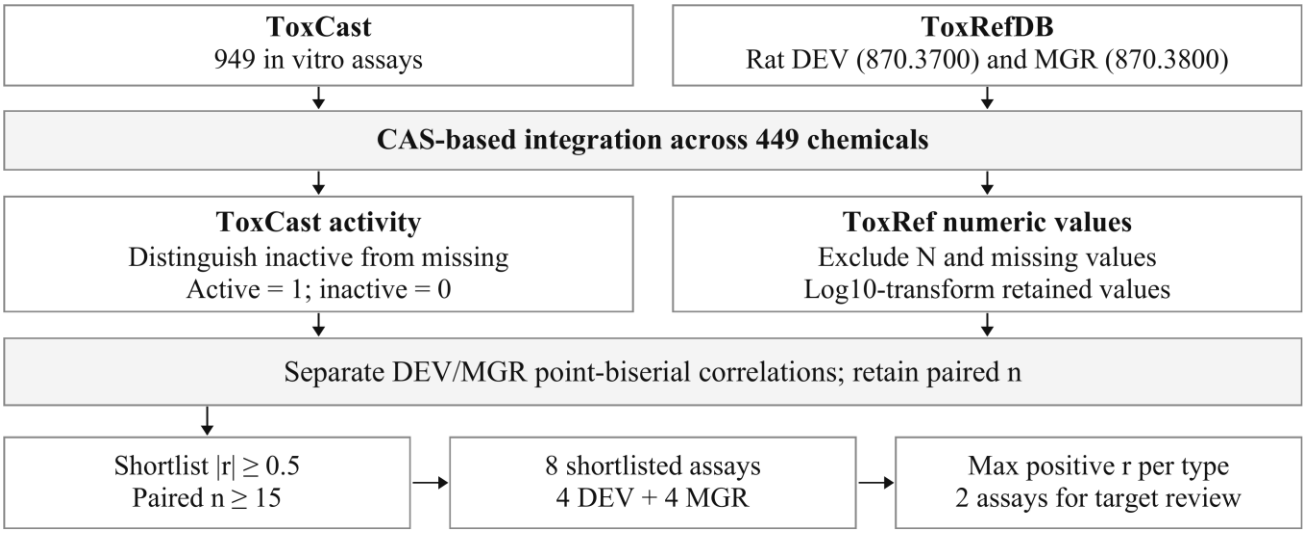


Figure 1. Data integration and assay prioritization. Preparation decisions lead to eight shortlisted associations and two assays selected for biological review.

Selected assay findings

Among the shortlisted assays, NVS_GPCR_rV1, annotated to Avpr1a, had the largest positive correlation with prenatal developmental study values ($r = 0.5677, n = 15$). NVS_GPCR_hM3, annotated to CHRM3, had the largest positive correlation with multigenerational reproductive study values ($r = 0.6245, n = 17$). Figure 2 shows all eight shortlisted assays, including negatively associated candidates. Positive coefficients indicate higher mean transformed dose-based values in the assay-active group. The sign alone does not establish toxicity severity.

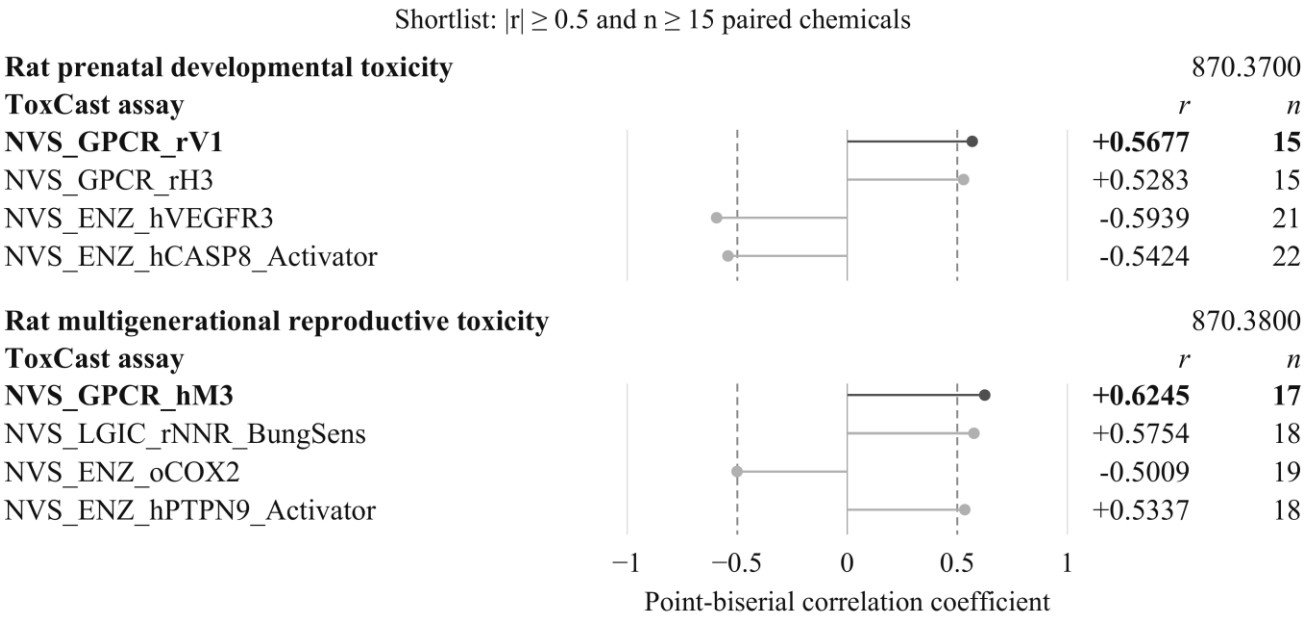


Figure 2. Shortlisted assay–endpoint associations. Point-biserial r relates assay activity (active = 1; inactive = 0) to $\log_{10}$-transformed ToxRef dose-based values; n is the paired chemical count. Dashed lines mark ±0.5. Dark gray marks the largest positive r per study type; a positive sign does not imply greater toxicity.

Biological interpretation

Both prioritized assays had targets in the rhodopsin-like G protein-coupled receptor family. The team reviewed UniProt and Comparative Toxicogenomics Database (CTD) information for AVPR1A and CHRM3. In the CTD information examined at the time, AVPR1A-associated entries included delayed effects of prenatal exposure and infertility; CHRM3 entries included infertility and fetal growth retardation. These database-based links provided context for follow-up hypotheses.

Findings and scope

The results support exploratory assay prioritization. The eight shortlisted associations used 15–22 paired chemicals; excluding ToxRef N entries restricted analysis to retained numeric outcomes. Correlations and database annotations do not establish causal mechanisms or screening accuracy. Independent chemical sets and evaluation against both positive and negative in vivo outcomes are needed to assess predictive performance.

Our findings were presented at the 2022 Research Workshop of the Graduate School of Chemical Safety Management, University of Seoul.